

Agent Hub: how I give my AI tools shared memory

I use a mix of Claude, Codex, and Grok depending on the task, and I'll often have them check each other's work. The problem with juggling tools like that is each one starts every session with a blank memory. Something I teach one tool rarely carries over to another, and keeping them in sync across a week of work meant constantly re-explaining the same context.

Agent Hub is how I fixed that. It's a single git repository that acts as shared long-term memory across every agent I use and every device I use them from. Any agent that can read files can read it. Any agent that can run git can write to it. Because it's a normal GitHub repo, it's also there on my phone, not just my laptop.

What it actually is

It's not an app, and it's not a service. It's a folder of markdown files, structured like a wiki, pushed to GitHub. That's the whole mechanism. The intelligence is a small set of conventions every agent gets told to follow, written down once in a file at the root.

The structure

```
agent-hub/
├─ AGENTS.md          # the contract: every agent reads this first
├─ CLAUDE.md          # symlink to AGENTS.md (Claude Code's expected
filename)
├─ HOME.md            # the index: start-here map of everything
├─ projects/          # one page per ongoing initiative, current state
only
├─ topics/            # cross-cutting concepts referenced by multiple
projects
├─ docs/              # finished artifacts (memos, specs, decision
docs)
├─ work/<project>/    # active working files, born here, committed as
they're made
│   └─ private/       # gitignored: sensitive stuff, local only,
never pushed
├─ log/               # append-only journal, one dated file per
completed task
```

```
└─ inbox/                # quick-capture drops (from mobile, usually),
waiting to be filed
└─ .claude/ .codex/ .cursor/ # each tool's native config, pointing at
shared conventions
```

Pages are wiki-style: short, and cross-linked with `[[double-bracket-links]]` (Obsidian's syntax, which is also how I browse the repo visually). A link resolves to a filename anywhere in the repo, so an agent can follow just the threads relevant to its task instead of reading everything.

The two protocols: this is the actual "how"

Reading, at the start of any task: 1. Read `HOME.md`. 2. Follow only the `[[links]]` relevant to the task at hand. 3. Don't bulk-read the whole repo. The link graph exists precisely so an agent loads cheap, targeted context instead of everything.

Writing, at the end of any meaningful task: 1. Log what happened to `log/YYYY-MM-DD-short-slug.md`: what was done, the key decisions and why, what was produced and where. 2. Update the relevant `projects/` page to reflect the new current state (the log holds history, the project page holds the present). 3. Link liberally. First mention of anything gets `[[linked]]`, even a page that doesn't exist yet (a "red link" just marks something worth writing later). 4. Update `HOME.md` if a new page got created. 5. `git add -A && git commit -m "log: <what happened>" && git push`.

Every agent, whichever tool it is, gets told to do the same read and write dance. That's what makes the memory shared instead of per-tool.

Staying honest about what's still true

Markdown drifts out of date, so pages tag individual facts with freshness labels instead of trusting one "last updated" date for the whole page:

- `Current as of YYYY-MM-DD`: for an active fact or plan
- `Decision (YYYY-MM-DD)`: for a settled choice, with the reason
- `Open question as of YYYY-MM-DD`: for something still unresolved, with an owner if I know one
- `Historical note (YYYY-MM-DD)`: for context that shouldn't drive current work

A label with only a start date tells you when something became true, not whether it still is. So a closed fact gets an end date and a pointer to what replaced it, instead of getting deleted or silently rewritten:

```
Decision (2026-06-01 → 2026-08-10, superseded by [[new-page]]):
```

A closed range means: don't act on this, follow the link for what's current now. An open range (start → present, or just a bare date) means: still active, but re-verify past 30 days. It's the same trick relational databases use for slowly changing history, a valid-from and a valid-to. It turns "is this still true?" from a judgment call into something an agent, or I, can resolve just by reading the label.

Keeping it tidy: the librarian

There's a small maintenance role, a "librarian," defined once and given to every tool (a Claude Code subagent, a Codex agent config, whatever the tool supports). On request ("maintain the hub"), it fixes broken links, files inbox items into the right pages, splits pages that have grown too long into a distilled current page plus a log entry, refreshes the `HOME.md` index, and flags stale projects. This is what keeps a wiki that several agents write into from turning into a junk drawer.

Keeping agents disciplined

"The agent will follow the rules" isn't a control, it's a hope. A written protocol only holds if something actually enforces it. Here's what does, in order of how much I trust it:

1. **There's no bulk-read affordance in the first place.** Reads go through individual file tools, one link at a time. There's no "dump the whole wiki" script or skill, and I won't build one later for convenience. That would undo the whole point.
2. **A stale or thin `HOME.md` is the real cause of over-reading,** not disobedience. An agent hedges and reads broadly when the index doesn't obviously answer "where do I look." Keeping `HOME.md` accurate is the load-bearing mechanism, not routine hygiene.
3. **The page-size cap is mechanical, not aspirational.** A pre-commit hook blocks committing any project or topic page over about 150 lines, so an oversized page can't slip into a commit unnoticed and turn up three months later at 800 lines.
4. **Runtime enforcement is the next level up, if I ever need it.** A tool hook that intercepts reads and warns or blocks on a real bulk-read pattern, or just counts files touched per turn. Worth building once I actually see the behavior, not pre-built against a hypothetical.
5. **Cost stays visible after the fact.** The log entry can note roughly how many pages a task touched, so a periodic pass can flag drift the same way it flags stale projects.

Privacy

Anything with personal data, customer data, or other sensitive material goes in a `private/` subfolder, which is gitignored: never pushed to GitHub, so it never reaches mobile or cloud agents either. It stays local, backed up however I back up my machine (not by GitHub). Everything else is plain markdown, safe to sync everywhere.

Cross-tool and mobile access

Each tool keeps its own native config format (Claude Code uses `.claude/`, Codex uses `.codex/`, Cursor uses `.cursor/`), but they all point at the same shared conventions and the same `AGENTS.md` contract. The instructions get written once, and every tool follows the same rules. Because the repo lives on GitHub:

- Claude on the web or on mobile can open it directly.
- Codex's cloud mode can check it out the same way.
- Any tool with a GitHub-aware agent mode can point at it the same way. You're not limited to your desktop.

One honest caveat: a tool needs some way to read and write GitHub (a native connector, or the ability to run git) to fully participate. A read-only tool can still consume the wiki, for instance by pasting `AGENTS.md` plus a project page into a chat, but it can't write back on its own.

What this costs, in tokens

Roughly nothing to run (it's markdown and git), but it isn't literally free. A task following the reading protocol pays for `AGENTS.md` plus `HOME.md` once (mine run around 4,200 tokens combined, at a rough four-characters-per-token estimate), plus whatever handful of linked pages the task actually needs (each capped around 150 lines, so maybe 500 to 800 tokens apiece, call it 1,500 to 3,000 more for two to four relevant links). So: roughly 5,000 to 8,000 tokens of added input per task, cold. That's the whole point of not bulk-reading the repo. Without that rule, you'd pay for the entire wiki every session instead of a slice of it, and the cost would grow as the wiki grows instead of staying roughly flat.

Two things soften this further. Prompt caching means repeat reads of `AGENTS.md` and `HOME.md` within the same session are billed at a fraction of full price. And the write side (a log entry plus a small page update) is only a few hundred tokens of generation; the git commit and push cost nothing, they're shell commands. The real lever on cost isn't the hub's existence, it's holding the line on the two policies above: keep pages small (now enforced by a hook, not just a convention), and don't let an agent read everything "to be safe."

Try it yourself

Pick whichever path fits how you work.

Option 1: clone the repo

I've published a working skeleton at github.com/eytaniam/public/tree/main/agent-hub-template. It already has the `AGENTS.md` contract, the folder structure, the pre-commit hook, and per-tool config examples in place. Grab it, run `git init`, push it as your own repo, and start writing.

Option 2: prompt an agent you already have

If you'd rather have your own Claude, Codex, or similar set this up for you from scratch, hand it this prompt as-is:

```
Set up a personal "agent hub": a git repo that serves as shared long-term memory across all the AI agents I use, so nothing I learn in one session is lost and I can access it from my phone too.
```

Create a new local folder and git repo with this structure:

```

AGENTS.md          – root contract file (see contents below)
CLAUDE.md          – symlink to AGENTS.md
HOME.md            – index page, frontmatter: type: topic, status:
active, updated: <today>
  projects/         – empty for now, one .md page per ongoing project
later
  topics/           – empty for now, cross-cutting concept pages later
  docs/             – empty for now, finished memos/specs go here
  work/             – empty for now; work/<project>/ folders are
created as work starts
  log/              – empty for now, append-only dated journal entries
  inbox/            – empty for now, quick-capture drops to be filed
later
.gitignore         – ignore: .DS_Store, work/*/private/,
work/private/, private/

```

Write AGENTS.md with this contract (adapt wording to taste but keep the mechanics):

1. Every page is a small wiki page: frontmatter (type, status, updated: YYYY-MM-DD) followed by content. Cross-reference other pages with `[[wiki-links]]` – a link resolves to a unique filename anywhere in the repo.
2. Reading protocol for any agent starting a task: read HOME.md first, then follow only the links relevant to the task. Never bulk-read the whole repo.
3. Writing protocol for any agent finishing a task: log the work to `log/YYYY-MM-DD-short-slug.md` (what happened, key decisions + why, what was produced and where); update the relevant projects/ page to reflect the new current state; link liberally, including to pages that don't exist yet; update HOME.md if a new page was created; then `git add -A && git commit -m "log: <summary>" && git push`.
4. Freshness labels on facts inside project pages, not just a page-level date:


```
"Current as of YYYY-MM-DD:", "Decision (YYYY-MM-DD):",
```

- "Open question as of YYYY-MM-DD:", "Historical note (YYYY-MM-DD):".
 When a fact is superseded or no longer true, close it with an end date and
 a pointer instead of deleting it, e.g.
 "Decision (2026-06-01 → 2026-08-10, superseded by [[new-page]]):".
 Treat anything unlabeled, open-ended, or >30 days old as needing re-verification.
5. Sensitive content (PII, customer data, anything you wouldn't want on GitHub)
 goes in a private/ subfolder inside the relevant work/<project>/ folder –
 gitignored, local-only, never pushed or synced to mobile/cloud agents.
6. Filenames: kebab-case, unique across the whole repo (this is what makes
 [[links]] unambiguous without a path).
7. Pages stay small – split anything that grows past ~150 lines into a
 distilled current-state page + a log entry holding the history. Make this
 mechanical, not aspirational: add a pre-commit hook (tracked in the repo,
 e.g. .github/hooks/pre-commit, enabled via `git config core.hooksPath .github/hooks` so it travels with clones) that fails the commit if any
 projects/*.md or topics/*.md file is over ~150 lines.
8. There is deliberately no "read the whole repo" script or skill –
 don't
 build one later for convenience. If HOME.md ever stops being a good
 enough index to answer "where do I look," that's the actual problem
 to
 fix, not a reason to add a bulk-read shortcut.

Then:

- Set up per-tool config so every agent I use (Claude Code, Codex, [others])
 reads AGENTS.md automatically at the start of a session, and follows the
 same read/write protocol above. Use each tool's native convention for this
 (e.g. Claude Code auto-reads CLAUDE.md at repo root).
- Create a GitHub repo and push this as the origin, so it's reachable from

mobile (Claude web/mobile, Codex cloud, or any other GitHub-aware agent).

- Optionally, define a small "librarian" helper (a subagent/config, whatever the tool supports) that on request: fixes broken `[[links]]`, files inbox/ items into the right pages, splits overgrown pages into log/ + distilled page, refreshes `HOME.md`, and flags projects untouched for 30+ days.

Confirm the structure with me before making the first commit, and ask which tools I actually use so you only wire up config for those.

Whichever path you take, three things are worth knowing going in:

- **The write step is what keeps it current, and nothing enforces it automatically.** Log, then update the page, then commit, at the end of every session. It's worth having your agent nag itself to do this rather than assuming it will.
- **Not every tool can write back equally well.** Anything with real git and file access can fully participate. A chat-only tool with no GitHub connection can still read the wiki if you paste in `AGENTS.md` plus a project page, but it can't log its own work back. That write has to come from a tool that can push.
- **Start small.** One `HOME.md`, one real project page, and the logging habit matter more than a fully built-out folder tree with nothing in it.

That's genuinely the system I run every day, across every machine I work from. If you set up your own version of it, I'd like to hear what you change.